# Zhenghe Guo

3230100923@zju.edu.cn
Academic Website Research Statement GitHub Google Scholar


Final-year undergraduate researcher working on embodied systems and spatial intelligence, with a focus on learned state representations, world modeling, and planning.

## Education

**Zhejiang University** Hangzhou, China
*B.Eng. in Agricultural Engineering; Minor in Advanced Class of Engineering Education* *2023–2027*
GPA: 3.86/4.00. Undergraduate research with Prof. Mingchuan Zhou.

**National University of Singapore** Singapore
*Visiting Student, Statistics and Data Science* *Jun–Jul 2025*
GPA: 4.00/4.00; decision-tree methods instructed by Prof. Wei-Yin Loh.

## Research Experience

**Westlake University** Hangzhou, China
*Visiting Research Intern, Inception3D, led by Prof. Anpei Chen* *May–Oct 2026*
Research on deformable-object manipulation and physics-grounded world models, combining MPM-based simulation with learned residual dynamics.

**Tsinghua University, Institute for AI Industry Research (AIR)** Beijing, China
*Visiting Research Intern, DISCOVER Lab* *Jan–Feb 2026*
Integrated a StereoHO-based perception baseline, calibration, visualization, and robot-side infrastructure for stereo hand-object reconstruction and handover experiments.

**Zhejiang University, Robotic Micro/Nano Manipulation Lab** Hangzhou, China
*Undergraduate Research Assistant, advised by Prof. Mingchuan Zhou* *Feb 2025–present*
Developing modeling, simulation, kinematics, and autonomous navigation methods for a magnetically controlled zebrafish robot.

**Zhejiang University, State Key Laboratory of CAD&CG** Hangzhou, China
*Undergraduate Research Assistant, advised by Prof. Ruizhen Hu* *Oct 2025–Jun 2026*
Research on instance-level reconstruction, with additional work on physics-grounded articulated-object reconstruction and street-scene reconstruction using COLMAP and FreeTimeGS.

**Zhejiang University, Agile Robotic Tele-systems (ARTs) Lab** Hangzhou, China
*Research Assistant, advised by Prof. Gaofeng Li* *Oct 2024–present*
Research on tactile sensing for biomimetic soft-tissue stress-field reconstruction, including sensor-layout optimization, active and sparse sampling, Gaussian-process inference, curved-surface tactile sensing, simulation, and data processing.

## Publications

**Zhenghe Guo**. "ObjectTwin: Measuring the Asset Correspondence Gap for Capture-Anchored Digital Twins." Submitted. [Project page]

Fanghao Wang*, **Zhenghe Guo***, Yihao Hu*, et al. "Spatio-Temporal Transformer Meta-RL for Robust Magnetic Microrobot Navigation in Cluttered Environments." Submitted.

**Zhenghe Guo**. "Sample-Efficient Residual Dynamics for Contact-Mismatched Deformable Simulation." Submitted. [Project page] [Report]

**Zhenghe Guo**. "Beyond State Prediction: Learning Action Consequences in Visual Physical Systems." Submitted. [Project page] [Report]

X. Yang, **Z. Guo**, S. Liang, Z. Yang, C. Zhu, and Z. Zhang. "Rate Maximization for UAV-assisted ISAC System with Fluid Antennas." *2025 IEEE/CIC International Conference on Communications in China Workshops*, pp. 1–5, 2025. [Paper]

**Zhenghe Guo** et al. "Action Identification and Validation Design for Prediction with Missing Covariates." Submitted to *Journal of Statistical Planning and Inference*. Manuscript available upon request.

*Equal contribution.

## Patents

Xiaoyan Chen, Xinyi Fang, Gaofeng Li, Qianming Chen, Zhichao Fan, **Zhenghe Guo**, Peng Deng, Misi Si, Rui Zhang. "Orthodontic Periodontal Ligament Stress Measurement Device and Stress Field Reconstruction Method." Chinese invention patent application. *Pending.*

Gaofeng Li, Zhichao Fan, Xiaoyan Chen, **Zhenghe Guo**, Xinyi Fang, Peng Cheng, Jiming Chen. "Method for Forming and Assembling a Simulated Periodontal Ligament Layer and Associated Testing Device." Chinese invention patent application. *Pending.*

Gaofeng Li, **Zhenghe Guo** (first student inventor), Xiaoyan Chen, Zhichao Fan, Xinyi Fang, Peng Cheng, Jiming Chen. "Physical Field Reconstruction Method and Device." Chinese invention patent application. *Pending.*

## Selected Projects

**DynaMo: Contact-Implicit Dynamic Motion Optimization and Energy-Aware Control for Humanoid Choreography** [Problem] [Paper] [Project page] 2025

- Designed a dynamics-aware framework combining contact scheduling, ZMP-aware centroidal control, whole-body coordination, and electrical-energy analysis; Asia and Pacific Mathematical Contest in Modeling (APMCM) National First Prize (top 2%).

**ZJUDancer: Vision & Localization for KidSize RoboCup Soccer** [Team] 2025–2026

- Developed and debugged ball, field, and robot perception, self-localization, and odometry integration for autonomous humanoid soccer; National First Prize and Runner-up at RoboCup China Open 2026.

**QUBO-Based Hybrid LDPC Decoding for Coherent Optical Optimization** 2025
[Problem] [Paper] [Project page]

- Engineered an LDPC simulation and decoding prototype with belief-propagation baselines, QUBO construction, adaptive penalties, partitioned solving, and BER/FER/runtime evaluation.

**RoboARM – Vision-Guided Gomoku Manipulation** [Code] 2025

- The RoboARM system autonomously plays Gomoku using YOLO/ArUco board perception, Negamax move selection, ROS/HTTP communication, and a calibrated six-axis arm with a pneumatic end effector.

**TurtleBot Autonomous Navigation** [Project page] 2025

- The TurtleBot stack integrates Nav2 with a custom ROS RRT* planner, using goal-biased sampling, KD-tree collision checks, obstacle inflation, rewiring, and real-robot path execution.

## Selected Awards

**Robotics**

- RoboCup China Open, Humanoid League (Small) – National First Prize, Runner-up, Team ZJUDancer, 2026.

**Communication and Innovation**

- Quantum Information Technology and Application Innovation Contest – National First Prize, Runner-up, 2025.
- China International College Students' Innovation Competition – Gold Medal, 2026.

**Mathematical Modeling**

- Asia and Pacific Mathematical Contest in Modeling (APMCM) – National First Prize (top 2%), 2025.
- DigitalCup Undergraduate Mathematical Contest in Modeling – National First Prize in Mathematics and Non-Mathematics groups, 2025.
- National Undergraduate Mathematics Competition, Non-Mathematics Category A – Second Prize, 2024.

**Scholarship**

- Xiangyang Scholarship, First Class (RMB 10,000; sole annual recipient in the college), 2024–2025.

## Technical Skills

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|---|---|
| **Programming** | Python, C/C++, MATLAB, R, JavaScript/TypeScript, Git, Docker, Linux |
| **Robotics** | ROS, OpenCV, PyTorch, CUDA, calibration, kinematics and dynamics, path planning, embedded systems, Arduino/STM32/Raspberry Pi |
| **Perception and ML** | 3D reconstruction, SfM, visual odometry, object detection, sensor fusion, reinforcement learning, optimization |

## Languages

Chinese (native) English (TOEFL 110)

Last updated October 2026. Detailed project reports, demos, and videos: https://styokuok.github.io/main/